

Topic: Review of Differential Equations

I. Introduction

Lets begin by considering the M th order differential equation of the form:

$$y^{(M)} + a_{M-1}y^{(M-1)} + a_{M-2}y^{(M-2)} + \cdots + a_1y' + a_0y = f(t) \quad (1)$$

where $M \in \mathbf{N}$; the coefficients $a_0, a_1, a_2, \dots, a_{M-2}, a_{M-1} \in \mathbf{R}$ and has the solution y (function of t).

Ex: Examples of this type of an equation would be:

$$y^{(5)} + 3y'' = \sin t \quad (2)$$

where $M = 5$, $a_4 = 0$, $a_3 = 0$, $a_2 = 3$, $a_1 = 0$, $a_0 = 0$, and $f(t) = \sin t$

$$3y''' + 6y'' - 14y = 0 \quad \implies \quad y''' + 2y'' - \frac{14}{3}y = 0 \quad (3)$$

where $M = 3$, $a_2 = 2$, $a_1 = 0$, $a_0 = -\frac{14}{3}$, and $f(t) = 0$

Defn: The differential equation in (1) is said to be **homogeneous** if $f(t) = 0$. It is called **non-homogeneous** otherwise.

Ex: Equation (2) is **non-homogeneous**

Equation (3) is **homogeneous**

Ex: Logically solve the following equations by filling in the blanks

$$(1) \quad 4y = 3t^2 : \quad 4\left(\frac{3}{4}t^2\right) = 3t^2 \quad \Rightarrow \quad y = \frac{3}{4}t^2 \quad \text{is a solution of (1)}$$

$$(2) \quad y' = 3t^2 : \quad \left(t^3\right)' = 3t^2 \quad \Rightarrow \quad y = t^3 + c \quad \text{is a solution of (2)}$$

*want these to cancel, so what
derivative looks like itself?*

$$(3) \quad y' + 4y = 0 : \quad \left(e^{-4t}\right)' + 4\left(e^{-4t}\right) = 0$$
$$-4e^{-4t} + 4e^{-4t} = 0$$
$$0 = 0 \quad \checkmark \quad \Rightarrow \quad y = e^{-4t} + c \quad \text{is a solution of (3)}$$

$$(4) \quad y' + 4y = 3t^2 : \quad \left(At^2 + Bt + C\right)' + 4\left(At^2 + Bt + C\right) = 3t^2$$

polynomial of deg 2

$$2At + B + 4At^2 + 4Bt + 4C = 3t^2$$
$$4At^2 + (2A + 4B)t + (B + 4C) = 3t^2 + 0t + 0$$
$$\Rightarrow \quad 4A = 3 \quad \Rightarrow \quad A = \frac{3}{4}$$
$$\Rightarrow \quad 2A + 4B = 0 \quad \Rightarrow \quad 2\left(\frac{3}{4}\right) + 4B = 0 \quad \Rightarrow \quad B = -\frac{3}{8}$$
$$\Rightarrow \quad B + 4C = 0 \quad \Rightarrow \quad \left(-\frac{3}{8}\right) + 4C = 0 \quad \Rightarrow \quad C = \frac{3}{32}$$
$$\Rightarrow \quad y = \frac{3}{4}t^2 - \frac{3}{8}t + \frac{3}{32} + c \quad \text{is a solution of (4)}$$

Important Question: Did we find all the solutions to the equations in the last example?

(1) $4y = 3t^2$: Is $y = \frac{3}{4}t^2$ the only solution? Yes

(2) $y' = 3t^2$: Is $y = t^3 + c$ the only solution? Yes

(3) $y' + 4y = 0$: Is $y = e^{-4t} + c$ the only solution? Yes

(4) $y' + 4y = 3t^2$: Is $y = \frac{3}{4}t^2 - \frac{3}{8}t + \frac{3}{32} + c$ the only solution? No. Why not?

Let $y_3 = e^{-4t} + c$ be the solution to Eqn 3 (i.e. $y_3' + 4y_3 = 0$) and

$y_4 = \frac{3}{4}t^2 - \frac{3}{8}t + \frac{3}{32} + c$ be the solution to Eqn 4 (i.e. $y_4' + 4y_4 = 3t^2$)

and consider the new function

$$y = y_3 + y_4 = e^{-4t} + \frac{3}{4}t^2 - \frac{3}{8}t + \frac{3}{32} + c$$

Plug y into Eqn (4):

$$\begin{aligned} y' + 4y &= \left[e^{-4t} + \frac{3}{4}t^2 - \frac{3}{8}t + \frac{3}{32} + c \right]' + 4 \left[e^{-4t} + \frac{3}{4}t^2 - \frac{3}{8}t + \frac{3}{32} + c \right] \\ &= \left[e^{-4t} \right]' + \left[\frac{3}{4}t^2 - \frac{3}{8}t + \frac{3}{32} + c \right]' + 4 \left[e^{-4t} \right] + 4 \left[\frac{3}{4}t^2 - \frac{3}{8}t + \frac{3}{32} + c \right] \\ &= \left[e^{-4t} \right]' + 4 \left[e^{-4t} \right] + \left[\frac{3}{4}t^2 - \frac{3}{8}t + \frac{3}{32} + c \right]' + 4 \left[\frac{3}{4}t^2 - \frac{3}{8}t + \frac{3}{32} + c \right] \\ &= 0 + 3t^2 = 3t^2 \quad \implies \quad y = y_3 + y_4 \quad \text{is also a solution of (4)} \end{aligned}$$

Follow-up Question: What does this tell us about solutions to the differential equation below?

$$y^{(M)} + a_{M-1}y^{(M-1)} + a_{M-2}y^{(M-2)} + \cdots + a_1y' + a_0y = f(t)$$

It says that the complete solution, $y(t)$, comes from the sum of solutions of the 2 equations below:

$$(1) \quad y^{(M)} + a_{M-1}y^{(M-1)} + \cdots + a_1y' + a_0y = 0 \quad (\text{homogeneous})$$

$$(2) \quad y^{(M)} + a_{M-1}y^{(M-1)} + \cdots + a_1y' + a_0y = f(t) \quad (\text{non-homogeneous})$$

Defn: The solution of eqn (1) is called the **homogeneous solution** denoted y_h and

The solution of eqn (2) is called the **particular solution** denoted y_p

Thm: The general solution, y , of

$$y^{(M)} + a_{M-1}y^{(M-1)} + a_{M-2}y^{(M-2)} + \cdots + a_1y' + a_0y = f(t)$$

is given by $y = y_h + y_p$

Notes: (1) The strategy we used to “guess” the form of the particular solution is known as the method of **“Undetermined Coefficients”**.

(2) The form of y_p we “guess” should not contain a constant multiple of any **function in y_h**

Ex: Find the solution of $y'' + 3y' - 4y = 4t + 5$; with $y(0) = 0$; $y'(0) = -1$; and given that

$$y_h(t) = c_1 e^t + c_2 e^{-4t}$$

Want: $y(t) = y_h(t) + y_p(t) = c_1 e^t + c_2 e^{-4t} + y_p(t)$ (y_h is given above)

Strategy: (1) Guess $y_p(t)$ with unknown coefficients (Look at the Diff Eqn)

(2) Compute $y_p'(t)$ & $y_p''(t)$

(3) Plug $y_p(t)$, $y_p'(t)$ & $y_p''(t)$ into the Diff Eqn

(4) Group the like terms

(5) Solve for the coefficients of $y_p(t)$

(6) Sub $y_p(t)$ into $y = c_1 e^t + c_2 e^{-4t} + y_p(t)$

(7) Apply the initial conditions to find c_1 & c_2

(1) Guess: $y_p(t) = At + B$

(2) Compute: $y_p'(t) = A$

$$y_p''(t) = 0$$

(3) Plug: $y_p'' + 3y_p' - 4y_p = 4t + 5$

$$0 + 3A - 4(At + B) = 4t + 5$$

(4) Group: $-4At + (3A - 4B) = 4t + 5$

(5) Solve: $-4A = 4$ & $3A - 4B = 5 \implies A = -1 \implies 3(-1) - 4B = 5 \implies B = -2$

$$y_p(t) = -t - 2$$

(6) Sub: $y(t) = c_1 e^t + c_2 e^{-4t} - t - 2$

$$y'(t) = c_1 e^t - 4c_2 e^{-4t} - 1$$

(7) Solve: $0 = y(0) = c_1 + c_2 - 2 \implies c_1 + c_2 = 2$

$$-1 = y'(0) = c_1 - 4c_2 - 1 \implies c_1 - 4c_2 = 0$$

$$c_1 = \frac{8}{5} \quad \& \quad c_2 = \frac{2}{5}$$

Therefore, $y(t) = \frac{8}{5}e^t + \frac{2}{5}e^{-4t} - t - 2$

What would the guess for y_p have been if we were instead solving $y'' + 3y' = 4t + 5$?

$$y_p(t) = At^2 + Bt + C$$

Ex: Find the particular solution of $y'' - 4y = 2e^{3t}$ given that we know $y_h = c_1e^{2t} + c_2e^{-2t}$

Guess: $y_p(t) = Ae^{3t}$

Compute: $y'_p(t) = 3Ae^{3t}$

$$y''_p(t) = 9Ae^{3t}$$

Plug: $y''_p - 4y_p = 2e^{3t}$

$$9Ae^{3t} - 4(Ae^{3t}) = 2e^{3t}$$

Group: $5Ae^{3t} = 2e^{3t}$

Solve: $5A = 2 \implies A = \frac{2}{5} \implies y_p(t) = \frac{2}{5}e^{3t}$

Ex: Find the particular solution of $y'' - 4y = 2e^{2t}$ given that we know $y_h = c_1e^{2t} + c_2e^{-2t}$

Guess: $y_p(t) = Ate^{2t}$ Why not Ae^{2t} ? Because e^{2t} is already a term in y_h

Compute: $y'_p(t) = Ae^{2t} + 2Ate^{2t}$

$$y''_p(t) = 2Ae^{2t} + 2Ae^{2t} + 4Ate^{2t} = 4Ae^{2t} + 4Ate^{2t}$$

Plug: $y''_p - 4y_p = 2e^{2t}$

$$4Ae^{2t} + 4Ate^{2t} - 4(Ate^{2t}) = 2e^{2t}$$

Group: $4Ae^{2t} = 2e^{2t}$

Solve: $4A = 2 \implies A = \frac{1}{2} \implies y_p(t) = \frac{1}{2}te^{2t}$

Ex: Find the solution of $3y'' + y' - 2y = 2 \cos t$; $y(0) = \frac{8}{13}$; $y'(0) = -\frac{14}{13}$; and $y_h = c_1 e^{\frac{2}{3}t} + c_2 e^{-t}$

Guess: $y_p(t) = A \cos t + B \sin t$

II. The Homogeneous Solution, y_h , for M th Order Linear DEs

Consider the M th order homogeneous differential equation

$$(1) \quad a_M y^{(M)} + a_{M-1} y^{(M-1)} + a_{M-2} y^{(M-2)} + \cdots + a_1 y' + a_0 y = 0$$

Defn: The M th degree polynomial

$$a_M r^M + a_{M-1} r^{M-1} + a_{M-2} r^{M-2} + \cdots + a_1 r + a_0 = 0$$

is called the **characteristic equation** of (1)

Thm: If $r_1, r_2, \dots, r_M$ are the roots of the characteristic equation then the solution of (1) is

$$y(t) = A_1 e^{r_1 t} + A_2 e^{r_2 t} + \cdots + A_{M-1} e^{r_{M-1} t} + A_M e^{r_M t}$$

where $A_1, A_2, \dots, A_{M-1}, A_M$ are constants.

Note: The roots of the characteristic equation can be complex.

III. The Homogeneous Solution, y_h , for 2nd Order Linear DEs

Consider the 2nd order homogeneous differential equation

$$(1) \quad a_2 y'' + a_1 y' + a_0 y = 0$$

which has the characteristic equation

$$a_2 r^2 + a_1 r + a_0 = 0$$

with roots: r_1, r_2

The possible forms of the solution to (1) can be summarized in the following table:

If	Then
$r_1 \neq r_2$ and are real	$y(t) = A_1 e^{r_1 t} + A_2 e^{r_2 t}$
$r_1 = r_2$ and are real	$y(t) = A_1 e^{r_1 t} + A_2 t e^{r_1 t} = (A_1 + A_2 t) e^{r_1 t}$
$r_1 \& r_2$ are complex (say $r_1 = a + ib$ & $r_2 = a - ib$)	$y(t) = e^{at} [A_1 \cos(bt) + A_2 \sin(bt)]$

Ex: Find the solution of $y'' + 4y = 3t^2$

Ex: Find the general solution of $3y'' + y' - 2y = 2 \cos t$

